$L_1 L_2$

$L_2 L_1 \rho$

$L_1 L_2$

$$\hat{\boldsymbol{\beta}} = {}_{\boldsymbol{\beta}} \left\{ \frac{1}{2n} \| - \boldsymbol{\beta}\|_2^2 + \alpha \left[\frac{1-\rho}{2} \|\boldsymbol{\beta}\|_2^2 + \rho \|\boldsymbol{\beta}\|_1 \right] \right\}$$

$\alpha \geq 0 \rho \in [0,1]$`l1_ratio`$L_1 L_2$

$\frac{1}{2n} \| - \boldsymbol{\beta}\|_2^2 \alpha \left[\frac{1-\rho}{2} \|\boldsymbol{\beta}\|_2^2 + \rho \|\boldsymbol{\beta}\|_1 \right] \alpha \rho L_1 L_2$

$\rho = 0 \equiv L_2$

$\rho = 1 \equiv L_1$

$0 < \rho < 1$

L_2

$x_1, \ldots, x_p$

$\boldsymbol{\beta}$

d

$p x_1, \ldots, x_p d$

$$\phi() = \left(x_1, x_2, \ldots, x_p,\ x_1^2, x_1 x_2, \ldots, x_p^2,\ \ldots,\ x_p^d \right)$$

$\binom{p+d}{d} - 1$

$p = 4$`total_femaletotal_malenight_mainlandnight_zanzibar`

$4 \to 14 x_i^2 x_i x_j$`night_mainland`^{2}

$4 \to 34 x_i^3 x_i^2 x_j x_i x_j x_k$

$d \binom{p+d}{d} L_1$

$(\alpha, \rho) \alpha \alpha \rho$

$K K K - 1 K K$

$$(\alpha^*, \rho^*) = {}_{\alpha,\rho} \frac{1}{K} \sum_{k=1}^{K} {}_k(\alpha, \rho)$$

$\alpha \in \{10^{-4}, 10^{-3.75}, \ldots, 10^{1}\}\alpha\alpha$
$\rho \in \{0.1, 0.3, 0.5, 0.7, 0.9, 0.95, 0.99\}\rho$
$50 \times 7 = 350350 \times 5 = 1750$

`elasticnet_polynomial.py`

`Train_new.csv`

$\rightarrow\rightarrow$
$y \leftarrow (1+)$

$\rightarrow$
$\alpha, \rho\rightarrow$
$d \in \{2, 3\}$
d
$d\rightarrow$
$d\rightarrow$

`total_femaletotal_male`> 13

`total_femaletotal_male`

`travel_withmost_impressing"Unknown"`

$\binom{185+2}{2} - 1 \approx 17{,}000\rightarrow$

$y = (1+)2.97{-}0.32y$

$(1+)$

R^2
$\alpha\rho$

$R^2 = 0.4639 R^2 = 0.3395$+0.124

$R^2 = 0.4642 \alpha^* = 0.0176 \rho^* = 0.70 R^2$`package_guided_tourpackage_insurance`

$\rho^* = 0.70$

$\alpha^* = 0.0176$

$\alpha p > n\alpha$

R^2

`tour_arrangementage_grouppayment_mode`

$\Rightarrow$

$R^2 = 0.13670.1216$

$\lvert\hat{\beta}\rvert$	
-0.1724	-0.1575

`tour_arrangement`0.3229

`night_mainland`0.2720`night_zanzibar`0.2513`night_mainlandnight_zanzibar`

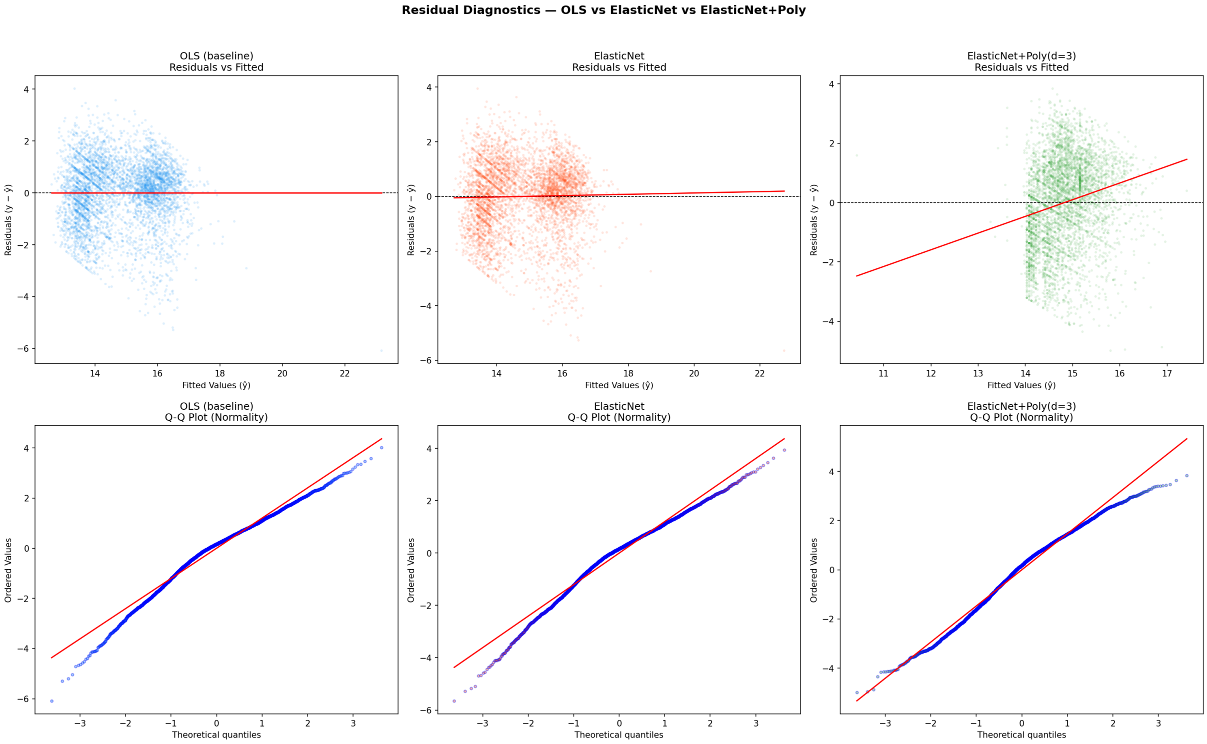
`purpose`-0.1575

$\mathbb{E}[\varepsilon|] = 0(\varepsilon|) = \sigma^2$

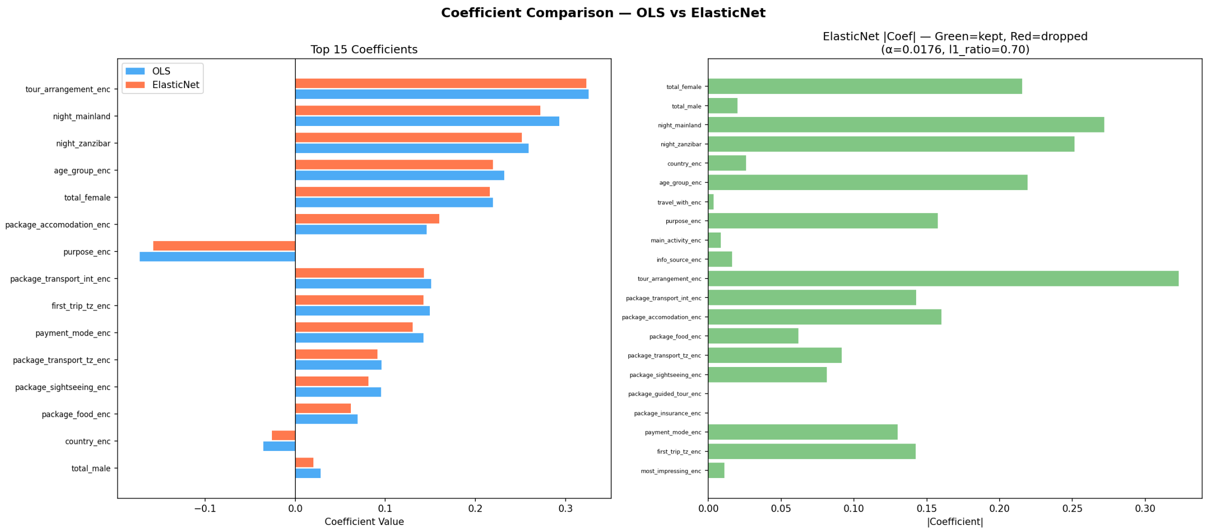
R^2

tF

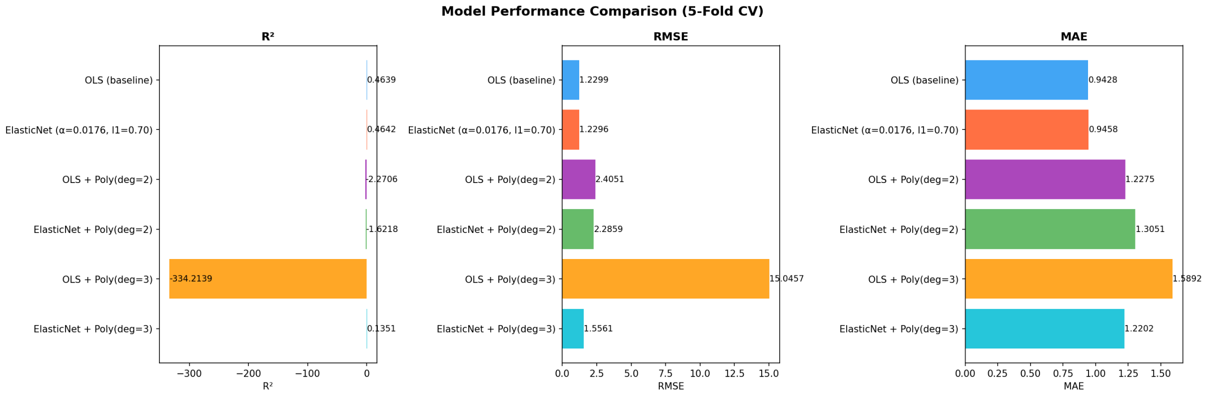
$n = 4{,}809$



$n > 4000$



$|\hat{\beta}||\hat{\beta}| > 0\hat{\beta} = 0$package_guided_tourpackage_insurance



R^2

$\alpha = 0.0176 \rho = 0.70 R^2 = 0.4642 R^2 = 0.3395 R^2 = 0.4639$`package_guided_tourpackage_insurance`

R^2`tour_arrangement`

`tour_arrangement`$>$`night_mainland`$>$`night_zanzibar`$>$`age_group`

`country`$\times$`tour_arrangementage_group`$\times$`main_activity`

`country`

$R^2 \approx 0.35 R^2 \approx 0.46 R^2 > 0.6$